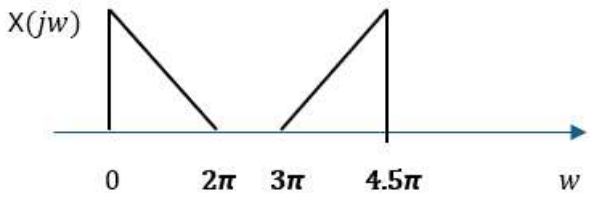
	SIGNALS AND SYSTEMS JULY-DEC 2025 ASSIGNMENT 4 Last Date for Submission: Nov 10 Answer Any 5 questions (write and upload)
1	What is the ROC of Z-transform function? With illustrative examples discuss the properties of ROC.
2	Explain the time shift property of Z-transform. Highlight the change in ROC due to time shifting. Illustrate with one example for (i) No change in ROC (ii) z = 0 gets excluded (iii) $\|z\| = \infty$ gets excluded.
3	Determine the Z-transform of the signal $x(n) = [n(-0.25)^n u(n)] * [(0.5)^{-n} u(-n)]$. Plot the poles and zeroes in z-plane and indicate ROC.
4	For the following Z-transforms, determine whether DTFT exists or not. If exists, determine DTFT. i) $H(z) = \frac{z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)(1+2z^{-1})}$; $\|z\| < \frac{1}{2}$ ii) $H(z) = \frac{z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)(1+2z^{-1})}$; $\frac{1}{2} < \|z\| < 2$
5	Determine the time domain signals for the H(z) using the different ROCs given below. if the ROCs are: $H(z) = \frac{1+\frac{7}{6}z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)\left(1+\frac{1}{3}z^{-1}\right)}$ if the ROCs are: i) $\|z\| > \frac{1}{2}$ ii) $\|z\| < \frac{1}{3}$ iii) $\frac{1}{3} < \|z\| < \frac{1}{2}$
6	Given: $x(t) = 4cos(8\pi t) + 6sin(12\pi t) + cos(20\pi t)$ Determine the minimum sampling frequency.
7	 Find the maximum sampling interval Ts.
8	Given the continuous-time signal: $x(t) = 3\ cos(6\pi t)$ Sketch the sampled $spectrum\ Xs(j\omega)\ for\ the\ following\ sampling\ intervals$: $(i)\ \ Ts = 1/8\ seconds$

	$(ii)\ Ts\ =\ 1/3\ seconds$ $(iii)\ Ts\ =\ 1\ second$ Show clearly whether aliasing occurs in each case.
9	Consider the continuous-time signal: $x(t)\ =\ 2cos(w_c t)$ The signal is sampled with sampling period: T_s = 1/4 seconds Plot the sampled spectrum X_s (jω) for the following values of w_c : (i) w_c = π (ii) w_c = 4π (iii) w_c = 5π Show clearly and indicate whether aliasing occurs in each case.
10	The spectrum of a signal $X(j\omega)$ is band-limited to $\mid \omega \mid \leq \omega_m$. The signal is sampled with period T_s . a) Write the expression for the sampled spectrum $X_s(j\omega)$. b) Determine the condition on T_s to avoid aliasing. c) Design the ideal reconstruction filter $H_r(j\omega)$ required to perfectly recover $X(j\omega)$.
11	A signal band-limited to $\mid \omega \mid \leq 500\pi$ rad/s is sampled with frequency $f_s\ =\ 600$ Hz a) Sketch $X_s(j\omega)$. b) Write the frequency response of the ideal reconstruction filter. c) State whether perfect reconstruction is possible and justify